Activity 3.5 – RISC-V Branch Unit

1.Design Approach:

Designed a RISC-V branch unit using Verilog HDL to support conditional branch instructions, JAL and JALR operations. Implemented comparison logic and branch target address generation.

2.Verification Steps:

- Created a separate testbench.
- Tested BEQ and BNE conditions.
- Verified signed BLT and BGE comparisons.
- Tested JAL instruction operation.
- Tested JALR target address generation.
- Checked branch target outputs.
- Simulated using ModelSim.

3.Results:

The testbench checks branch decisions and target address generation for the implemented branch operations. PASS/FAIL messages are displayed.

4.Issues Faced:

Signed comparisons must be implemented correctly. Branch target addresses require proper offset addition. JALR address alignment must be handled correctly.

5.Conclusion:

Successfully designed and verified a RISC-V branch unit using Verilog HDL.